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Product Name / SKU

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Unit Heart

SKU:U029



Description

Unit Heart is a blood oxygen and heart rate sensor. Integrated **MAX30100**, it offers a complete pulse oximeter and heart rate sensor system solution. This is a non-invasive blood oxygen and heart rate sensor, integrating two infrared light-emitting diodes and a photodetector. The detection principle is using infrared LEDs to illuminate and detect the proportion of red blood cells carrying oxygen versus those not carrying oxygen, thereby obtaining the oxygen content.

Test method: After the program runs, place your finger in the detection area.

Heart Unit communicates with the Core via the I2C protocol, I2C address: 0x57

Features

- Programmable sample rate and LED current to save power consumption
- Ultra-low shutdown current ($0.7\mu\text{A}$ typical)
- High measurement performance
- High sampling rate
- Built-in temperature sensor calibration
- Fast data output
- Blood oxygen concentration detection, heart rate detection
- 2 x LEGO compatible holes
- Development Platform: Arduino

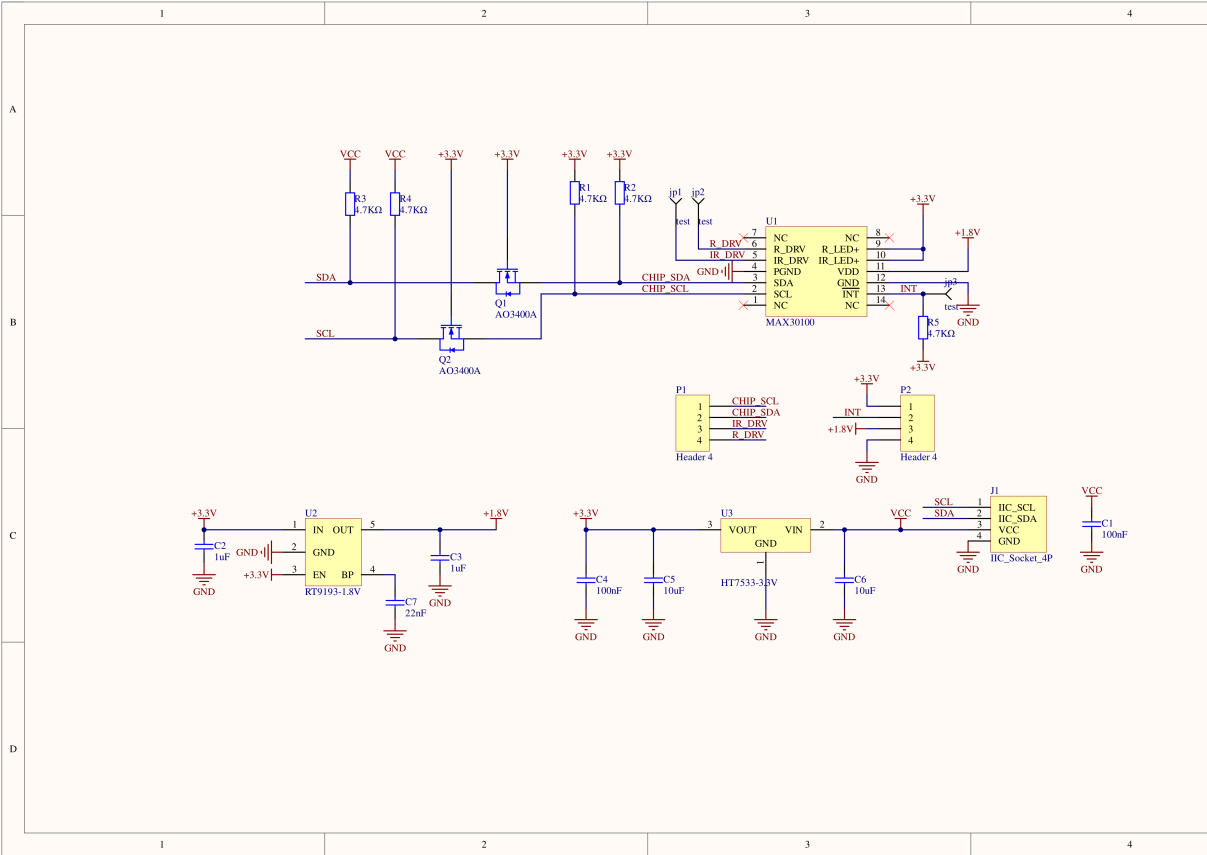
Includes

- 1 x Unit Heart
- 1 x HY2.0-4P Grove Cable (20cm)

Specifications

Specification	Parameter
Protocol	I2C:0x57
Product Size	32.0 x 24.0 x 8.0mm
Product Weight	4.9g
Package Size	138.0 x 93.0 x 9.0mm
Gross Weight	10.0g

Schematics

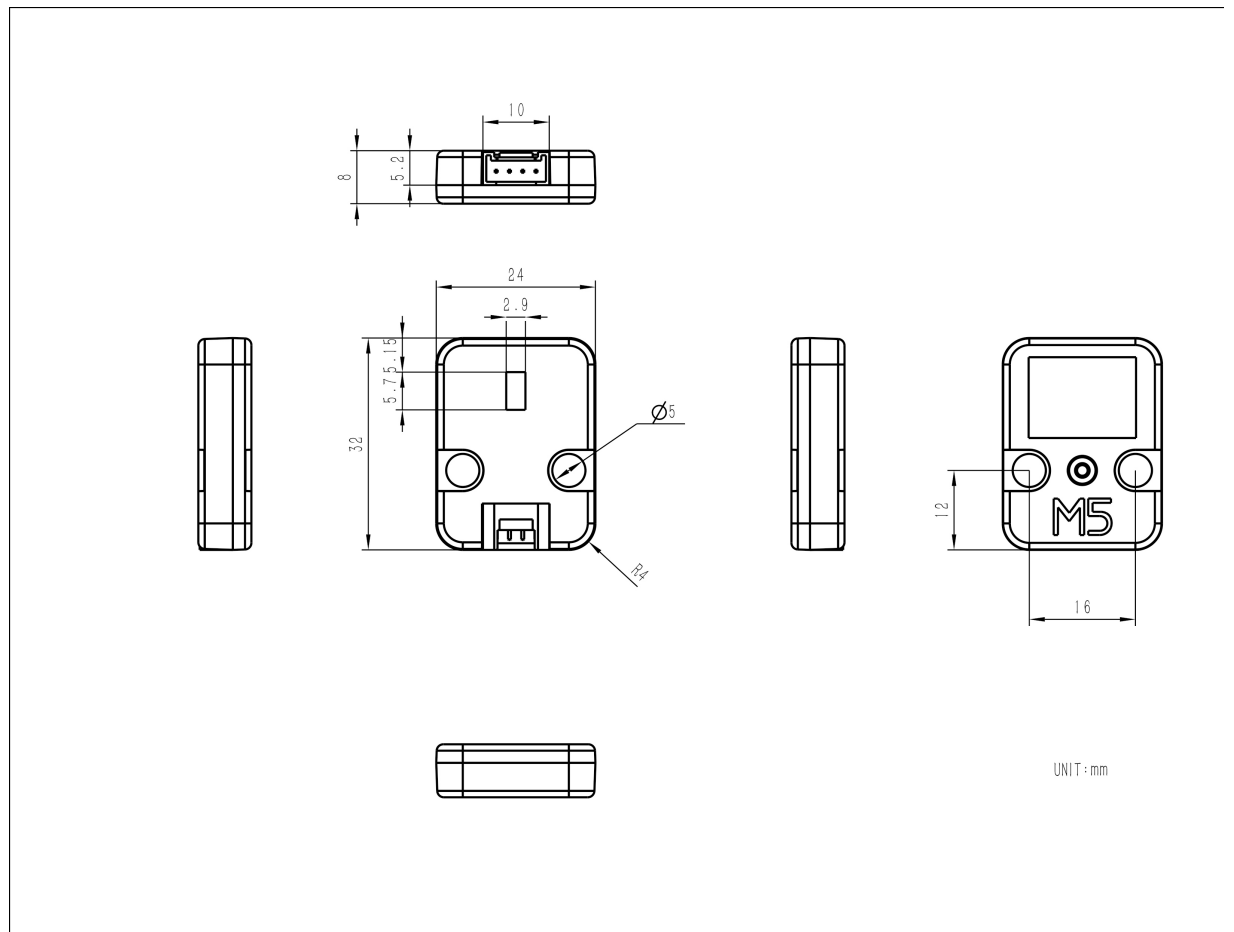


PinMap

HY2.0-4P

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

Model Size



Datasheets

- [MAX30100](#)
- [MAX30100 lib](#)

Softwares

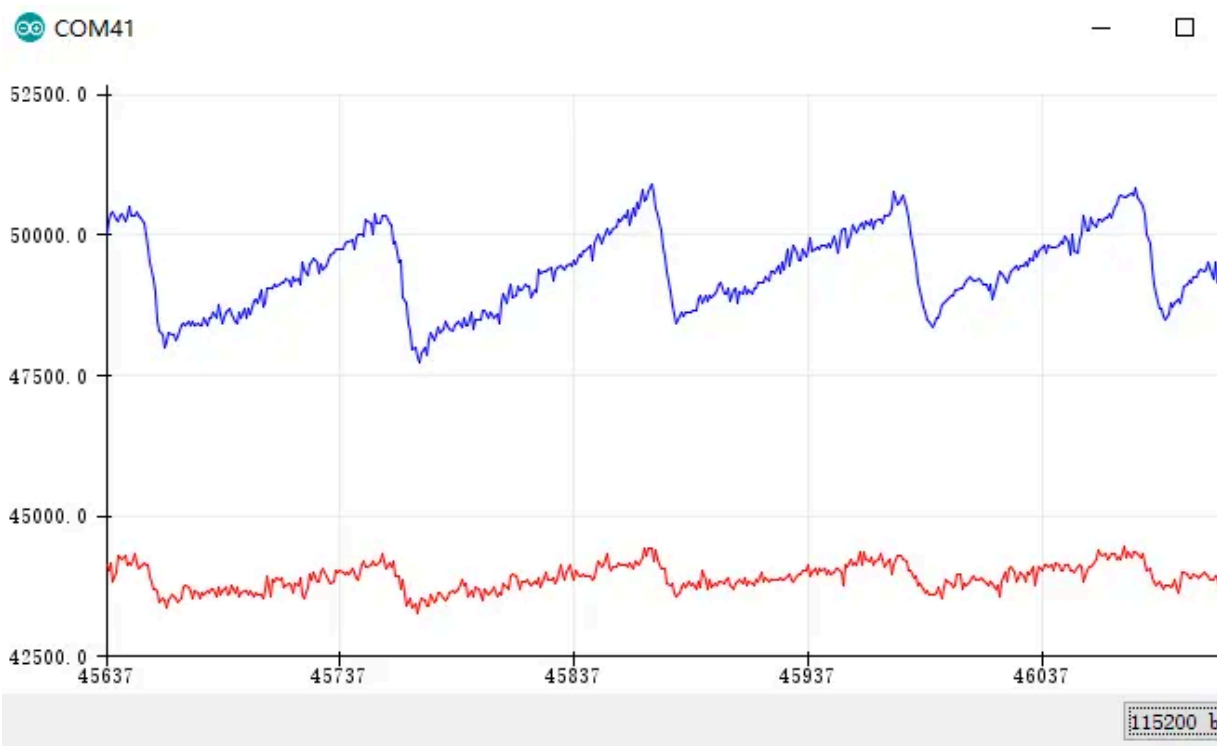
Arduino

- [Unit Heart Test Program](#)



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- [Unit Heart Arduino Quick Start](#)

UiFlow1

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UiFlow2

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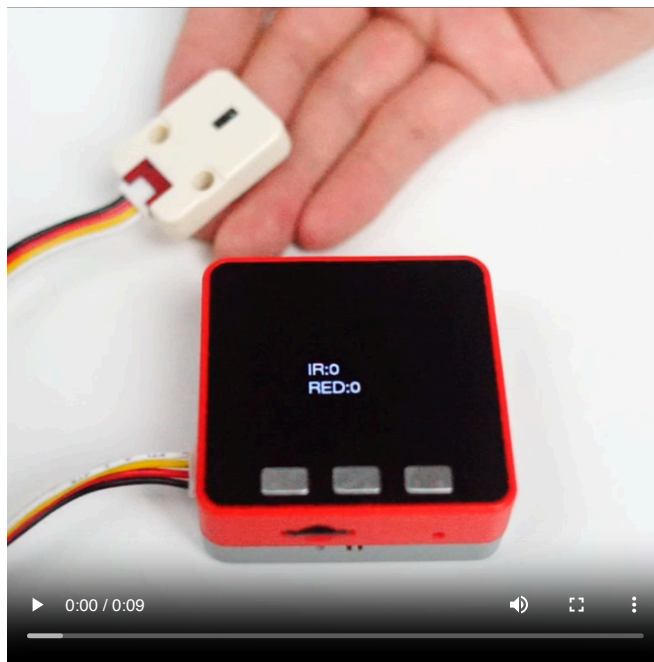
Easyloader

Easyloader	Download	Note
Unit Heart Test Easyloader	download	

Video



- Display blood oxygen and heart rate sensor detection values on the screen



Bilibili

Youtube



UiFlow2 Unit

Unit Heart | ...



00:01 / 02:24



| Compatible



Unit Heart

Unit-PORT

PORT.A			
GND	5V	I2C_SDA	I2C_SCL

+



CoreS3

Available PORT

PORT.A			
GND	5V	G2	G1

PORT.B			
GND	5V	G9	G8

PORT.C			
GND	5V	G17	G18

"Unit Heart" & "CoreS3" Grove Port Connection Note:

Please modify the example program according to the actual PIN number when using.

Available Ports: [PORT.A](#) [PORT.B](#) [PORT.C](#) .

| FAQ

Q1: What is the accuracy and resolution of the Heart rate unit?

